

Towards Compression-Aware Geospatial Architecture using JPEG AI Latents for Road Segmentation

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Abstract—The imagery captured by modern Earth-observation satellites far outmatches their downlink capacity, so imagery must be compressed before it can be returned to the ground. Learned image codecs such as JPEG AI compress satellite imagery into structured latent tensors before transmission; conventional analysis pipelines reconstruct the full image before applying any downstream task. We hypothesise that the dominant source of degradation in compressed-image segmentation is the synthesis decoder itself, rather than latent quantisation, and that operating directly on entropy-decoded latents avoids this artefact while sidestepping the decode step entirely. We present LILNet (4.3 M parameters) and BigLILNet (14.4 M), UNet-style encoder-decoders trained jointly across the four JPEG AI quality levels for bitrate-agnostic road segmentation. On a combined three-source benchmark (Massachusetts Roads, DeepGlobe, OpenEarthMap; 2,065 test tiles) evaluated over six compression ratios, BigLILNet attains the best IoU and F1 of any method across the high-compression regime (32:1–200:1), outperforming five RGB-domain baselines, including road-specific networks; at 200:1 it exceeds the strongest baseline by 0.084 IoU. A crossover near 24:1–32:1 marks the point at which full-resolution detail begins to outweigh the latent’s artefact immunity, providing a practical criterion for pipeline selection under a transmission budget. Using up to an order of magnitude fewer parameters and eliminating the synthesis decoder from the inference path, the approach enables onboard road extraction directly on the compressed-domain representation.

Index Terms—JPEG AI, learned image coding, road segmentation, latent-domain inference, remote sensing

I. INTRODUCTION

Modern Earth observation satellites generate data at volumes that strain conventional ground-based pipelines: the Copernicus Sentinel-2 mission alone downlinks on the order of 1.7 TB of imagery per day [1]. Sensors have grown from a handful of coarse multispectral bands to instruments capturing dozens or hundreds of spectral channels at increasingly fine spatial resolutions [2], and constellation sizes have expanded from single spacecraft to hundreds of satellites simultaneously imaging [3]. The compounding effect on data volumes makes

onboard compression an increasingly recommended strategy as the demand for timely delivery tightens [4]. In the conventional pipeline, this compressed imagery is downlinked and then fully reconstructed on the ground before any analysis runs, even when the end product is a compact map such as a road network. As learned image coding matures and standards such as JPEG AI [5] offer state-of-the-art compression efficiency, the question arises: can inference run directly on the compressed representation, eliminating the decode step and enabling lightweight analysis onboard the satellite?

Prior work has shown that latent-domain inference on JPEG AI tensors is viable for visual recognition: Alkhateeb et al. [6], [7] demonstrate this for face detection, where latent tensors, free of pixel-domain artefacts, provide a compact structured encoding that downstream models can exploit directly. Yet this remains limited to image-level recognition, with no latent-domain method demonstrated for a geospatial, remote-sensing task. Segmentation and, in particular, road extraction are far more demanding: thin road networks, ambiguous junctions, and high intra-class variation require precise boundary localisation, which reconstruction artefacts specifically undermine [8]. At high compression, blurring and ringing introduced during image reconstruction degrade boundary sharpness and fine spatial structure [9], implicating the decode step rather than the encoded representation as the primary source of degradation.

We hypothesise that operating directly on JPEG AI latent tensors offers two complementary advantages over the conventional decode-then-infer pipeline: Bypassing image reconstruction eliminates the blurring and ringing artefacts it introduces, producing a representation free of decoder-induced distortions across all compression levels. The analysis transform performed by the JPEG AI encoder is itself a substantial feature-extraction step; a downstream model operating on the resulting latents avoids repeating this computation, reducing per-tile processing costs and enabling more efficient inference directly from the

compressed bitstream. We realise this as two lightweight latent-domain networks, LILNet and BigLILNet, and evaluate them against pixel-domain baselines across the full range of JPEG AI compression levels on a multi-source road benchmark.

To our knowledge, this is the first work to perform dense, pixel-level prediction directly on JPEG AI latents, and the first to apply compressed-domain inference in a geospatial, remote-sensing setting. Our primary contributions are:

- LILNet, a lightweight image latent (LIL) encoder–decoder for compression-robust road segmentation directly from JPEG AI latent tensors, without requiring image reconstruction.
- BigLILNet, a higher-capacity variant that replaces the bottleneck with an atrous spatial pyramid pooling block, testing whether latent-domain accuracy scales with model capacity.
- A systematic evaluation of latent-domain inference against pixel-domain baselines across six compression levels on a combined three-source road segmentation benchmark.

II. RELATED WORK

Learned image coding. Onboard Earth-observation compression has historically relied on classical still-image standards, most notably the wavelet-based JPEG 2000 [10] and the CCSDS image-compression recommendation [11]. Learned codecs replace the hand-designed transforms of classical standards with analysis and synthesis networks trained end-to-end for rate–distortion [12]; hyperprior and autoregressive entropy models [13], [14] soon overtook the best classical codecs and carried the field from research prototype to JPEG AI [5], the first learned-coding standard. What matters here is that JPEG AI does not merely compress: its analysis transform yields a structured latent that is already rich in visual features. This representation, and the feature extraction that produced it, is normally discarded the instant the image is reconstructed. Even methods built to compress features for satellite downlink [4] reconstruct before they analyse.

Compressed-domain inference. A growing body of work treats reconstruction as avoidable rather than a prerequisite for analysis. Convolutional networks have been trained directly on classical block-DCT coefficients [15], and Torfason et al. [16] even performed semantic segmentation on a learned codec’s latent space without decoding to pixels. On JPEG AI itself, however, only image-level recognition has been shown, namely face detection [6], [7]. Whether its latent retains the fine spatial structure a dense prediction task demands is therefore open. Since compression artefacts primarily harm downstream tasks at the reconstruction stage [9], a boundary-critical task is the sharpest test of this.

Road extraction. Roads are among the thinnest, most fragmentation-prone targets in remote sensing [8], where a few misclassified pixels can sever a whole connected route. General encoder–decoders such as UNet [17], SegNet [18], and DeepLabV3+ [19], and a lineage of connectivity- and topology-aware road networks (D-LinkNet [20], the residual

UNet of Zhang et al. [21], and the recent SegRoadv2 [22] and TopoRF-Net [23]) have pushed accuracy steadily higher, yet all share one assumption: they operate on reconstructed pixels, and so inherit precisely the artefacts that compression introduces. This makes road extraction an ideal stress test for latent-domain inference, and the gap this paper closes.

III. METHODOLOGY

We propose two latent-domain encoder–decoders that map a JPEG AI latent directly to a road mask without reconstructing the image: LILNet (Fig. 1a), a lightweight segmentation model, and BigLILNet (Fig. 1b), which trades size for capacity to test how far latent-domain accuracy scales. Both are built to the same constraints: a single set of weights spanning the codec’s quality levels, robustness to the sparse, uneven latents those levels produce, and a footprint small enough for onboard use.

A. Network Architectures

JPEG AI [5] encodes a 1024×1024 input into a $256 \times 64 \times 64$ latent: a $16 \times$ spatial downsampling, with 256 channels that concatenate the luminance and chrominance components. Channel occupancy varies with compression: at low BPP, aggressive quantisation zeros a large fraction of channels, so a standard convolutional front end wastes capacity on uninformative input. Both models address this with a channel-adaptive input front end (Fig. 1).

LILNet (Fig. 1a) is a UNet-style encoder–decoder that maps the latent $\hat{y}$ from the JPEG AI bitstream to a 1024×1024 binary logit. The latent is first normalised with GroupNorm rather than BatchNorm, since tile-wise deployment runs at batch size 1, where batch statistics are unreliable. A Squeeze-and-Excitation block [24] then recalibrates the channels, emphasising the informative components and suppressing the near-zero channels that dominate at low BPP. A transposed convolution upsamples this representation from 64×64 to 128×128 , after which three residual double-convolution stages with max-pooling contract the feature map to a 16×16 bottleneck; the 1×1 residual projections preserve gradient flow when activations are sparse at low BPP.

The decoder restores resolution via transposed convolutions, with skip connections linking the three encoder stages to their corresponding decoder stages, thereby reinstating spatial detail lost to pooling. Three further upsampling stages take the feature map to full 1024×1024 resolution, and a 1×1 convolution maps the feature channels to the per-pixel road logit. LILNet totals 4.3 M parameters.

BigLILNet (Fig. 1b) tests whether latent-domain performance scales with capacity or is bounded by the compressed representation itself. It doubles the LILNet encoder width to $C_{\text{enc}} = 256$ and replaces the residual-convolution bottleneck with an Atrous Spatial Pyramid Pooling block (Fig. 1b), whose parallel dilated branches (rates 2 and 4, matched to the 16×16 bottleneck) aggregate multi-scale context where each cell already spans a 64×64 pixel region, helping retain thin roads that occupy only a few cells. It otherwise shares LILNet’s depth and structure, totalling 14.4 M parameters ($3.3 \times$ LILNet).

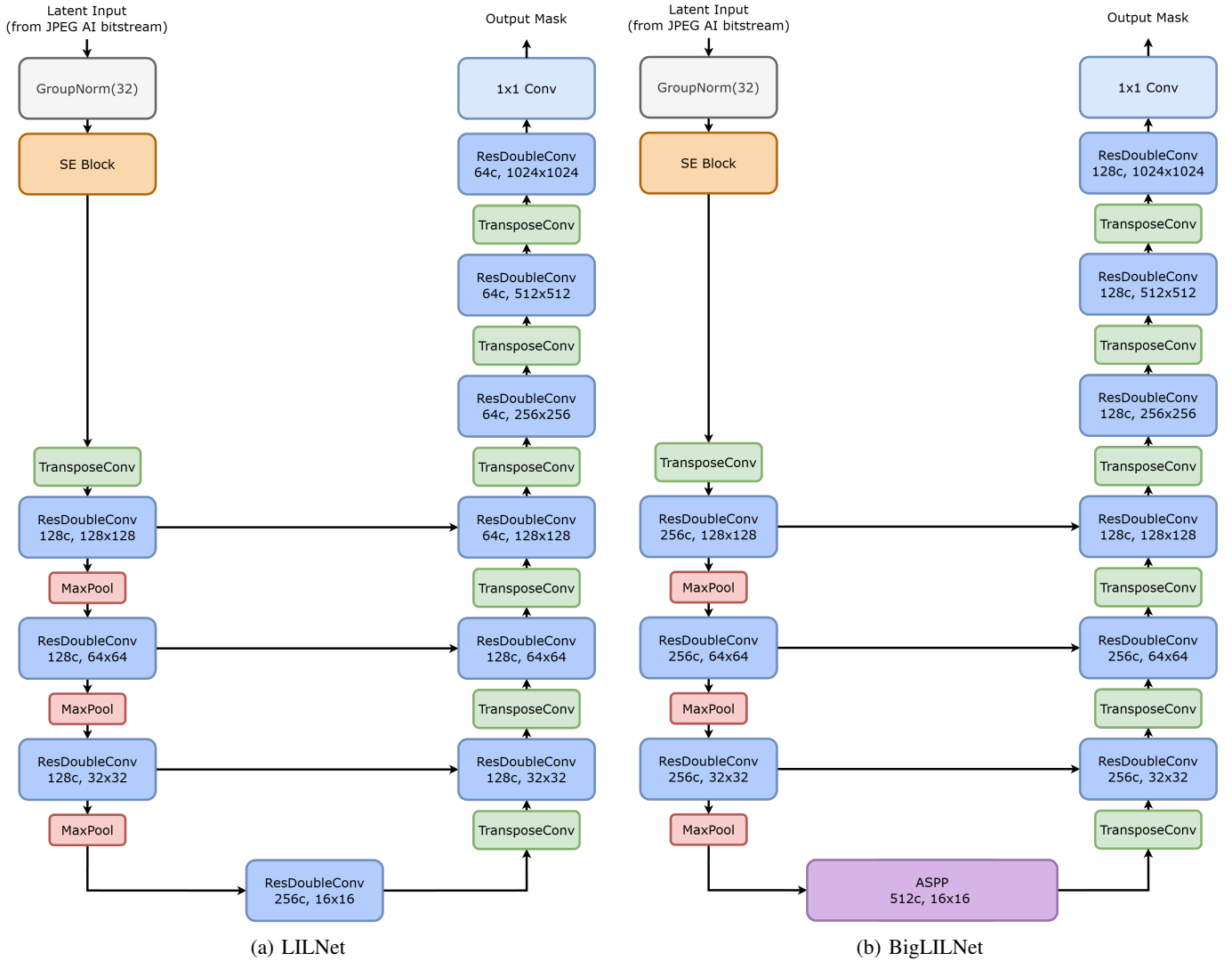


Fig. 1: Architecture of the two proposed networks. Both share a channel-adaptive input front end (GroupNorm followed by a Squeeze-and-Excitation block that reweights the sparse latent channels) and a transposed-convolution bridge that upsamples the 64×64 latent, feeding a three-stage residual encoder–decoder with skip connections. (a) LILNet (4.3,M) uses a residual-convolution bottleneck; (b) BigLILNet (14.4,M) doubles the encoder width and swaps the bottleneck for an Atrous Spatial Pyramid Pooling (ASPP) block for multi-scale context.

B. Training

Both models are trained with Dice loss, which normalises the objective over the foreground class and is intrinsically robust to the severe class imbalance of road segmentation, where roads occupy roughly 5% of pixels [25], without explicit positive-class weighting [21].

JPEG AI defines four quality levels, each a separately trained codec with its own quantisation regime, rather than a bitrate setting for a shared model. A segmentation network trained at a single level would encounter a shifted latent distribution when deployed at another level, so both models are trained across all four levels, resampling the target level uniformly at each iteration. Gaussian latent noise ($\alpha=0.02$) further discourages fitting the quantisation grid of any single codec.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

1) *Datasets:* We train, validate, and test on a combined, multi-source benchmark comprising three publicly available remote-sensing road datasets. Massachusetts Roads [25] provides aerial imagery of urban and rural Massachusetts at approximately 1 m GSD. DeepGlobe Road Extraction [26] provides satellite imagery spanning diverse geographies and urban typologies. OpenEarthMap (OEM) [27] contributes globally distributed imagery with road annotations derived from a multi-class land-cover scheme. The combination provides the geographic and radiometric diversity needed to reduce overfitting to the annotation style or spatial statistics of any single source.

TABLE I: Road extraction mean performance (IoU / F1) against various compression ratios (CR) on JPEG AI-encoded remote sensing imagery. Best per column in **bold**, second best underlined.

Method	8:1		24:1		32:1		48:1		96:1		200:1		Params (M)	ms/ tile
	CR		CR		CR		CR		CR		CR			
UNet [17]	0.622	0.767	0.606	0.754	0.481	0.650	0.468	0.638	0.411	0.583	0.221	0.362	31.4	55.6
SegNet [18]	0.599	0.749	0.587	0.740	0.499	0.666	0.489	0.657	0.444	0.615	0.299	0.461	29.5	<u>32.0</u>
DeepLabV3+ [19]	0.577	0.732	0.572	0.728	0.499	0.666	0.489	0.657	0.450	0.621	0.302	0.463	39.7	58.8
SegRoadv2 [22]	<u>0.600</u>	<u>0.750</u>	<u>0.593</u>	<u>0.744</u>	<u>0.507</u>	<u>0.673</u>	<u>0.498</u>	<u>0.665</u>	<u>0.458</u>	<u>0.628</u>	0.352	0.521	25.9	149.2
TopoRF-Net [23]	0.554	0.713	0.547	0.707	0.457	0.627	0.446	0.617	0.409	0.580	0.312	0.475	28.4	176.4
LILNet (ours)	0.498	0.665	0.492	0.659	0.485	0.653	0.476	0.645	0.452	0.623	<u>0.396</u>	<u>0.567</u>	4.3	20.2
BigLILNet (ours)	0.547	0.708	0.527	0.691	0.515	0.680	0.509	0.674	0.469	0.639	0.437	0.608	<u>14.4</u>	34.7

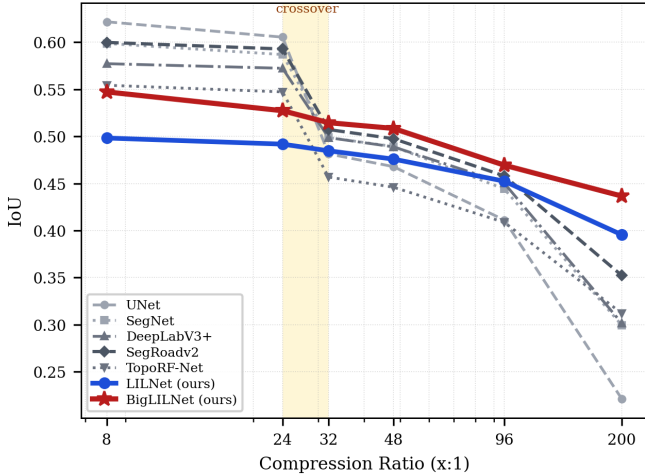


Fig. 2: Segmentation IoU versus compression ratio, highlighting the compression ratio crossover between 24:1 and 32:1 where BigLILNet surpasses all baseline models’ IoU performance.

All images are resized or tiled to 1024×1024 pixels, yielding 64×64 latent tensors after JPEG AI encoding. The combined dataset is split at the image level into 17,255 training, 2,132 validation, and 2,065 test tiles, with no tile-level overlap across splits, ensuring evaluation is free of spatial leakage. For latent-domain training, all images are pre-encoded with the JPEG AI reference software using each of the four quality-level codecs and the resulting latent tensors stored to disk prior to training; this decouples codec execution from the training loop and permits efficient multi-worker data loading.

2) *Baselines*: We compare against five RGB-domain baselines, each trained on the uncompressed imagery of the same 17,255-tile training set used for the latent-domain models: UNet [17], SegNet [18], DeepLabV3+ [19], SegRoadv2 [22], and TopoRF-Net [23]. The first three are general-purpose segmentation networks; SegRoadv2 and TopoRF-Net are recent road-specific networks that add explicit connectivity and topology modelling to better preserve continuous road structure. At test time, each RGB baseline receives JPEG-

AI-decoded imagery at the evaluation compression level (the conventional decode-then-infer pipeline), whereas our latent-domain models receive the corresponding entropy-decoded latent tensors directly.

3) *Implementation Details*: Both models are implemented in PyTorch and trained using the same recipe, differing only in architecture. Training runs for 50 epochs with the AdamW optimiser [28], a cosine-annealed learning rate schedule (peak $\eta = 1 \times 10^{-4}$, linear warmup over 2 epochs, minimum $\eta = 1 \times 10^{-6}$), and weight decay 2×10^{-4} . AdamW is chosen because its per-parameter adaptive step sizes accommodate the large variation in channel-magnitude scale that JPEG-AI latents exhibit across compression levels; a uniform learning rate would under-step high-magnitude channels and over-step the near-zero channels dominant at low BPP. Effective batch size 8 is achieved via gradient accumulation (physical batch size 2 over 4 accumulation steps) to fit within the 8 GB VRAM of the training platform (NVIDIA RTX 2060 SUPER). Dropout (rate 0.25) is applied at the bottleneck only.

4) *Evaluation Metrics*: Segmentation quality is reported as Intersection-over-Union (IoU) and F1 score (Dice coefficient) at each of six evaluation compression levels. Four levels (0.12, 0.50, 1.00, 3.00 BPP) correspond to the four JPEG AI quality-level codecs seen during training; two additional levels (0.25 and 0.75 BPP) are held-out interpolation points that probe generalisation to compression rates falling between the trained codecs. Per-tile inference latency (ms) is measured at batch size 1 on the RTX 2060 SUPER to reflect single-tile deployment. The reported figure is the model forward pass only: for the latent-domain models, it is run on an entropy-decoded latent tensor, and for the baselines on a fully decoded 1024×1024 image. Entropy decoding is excluded from both pipelines, as it is common to both. The reported numbers, therefore, isolate the cost of the segmentation network and, if anything, understate the latent-domain advantage: the decode-then-infer pipeline additionally incurs the JPEG AI synthesis transform that the latent-domain models skip entirely, a cost not charged against the baselines here.

Table I and Fig. 2 establish the central result: across the high-compression regime that matters most for constrained down-

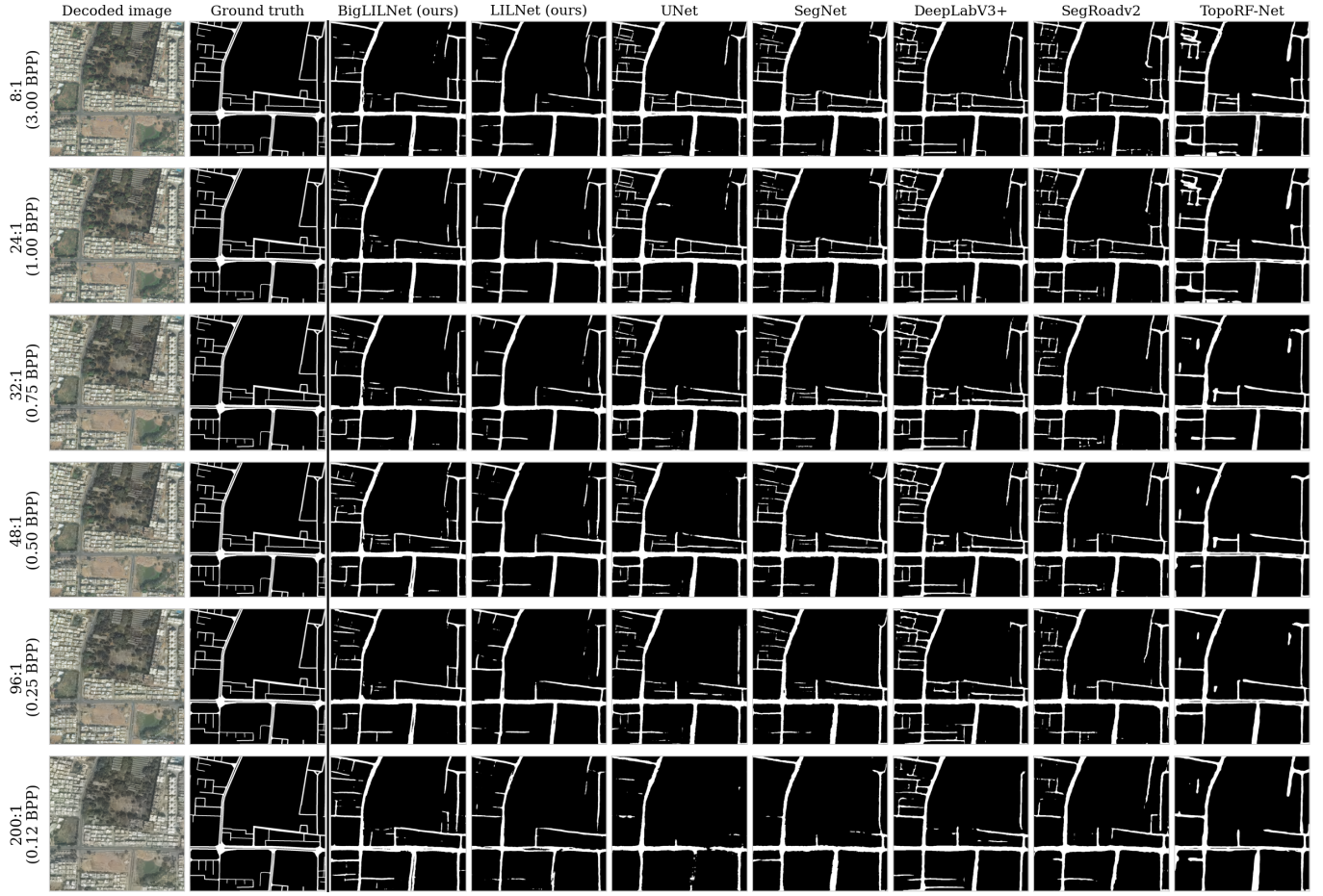


Fig. 3: Qualitative road extraction across the full compression range on a held-out tile. Rows are compression ratios from 8:1 (3.00 BPP, top) to 200:1 (0.12 BPP, bottom). The two leftmost columns give the JPEG AI-decoded image at each rate (its degradation visible from top to bottom) and the fixed ground-truth mask; a rule separates these references from each method’s predicted mask. As the decoded image degrades, the latent-domain models retain connected road structure where the pixel-domain baselines fragment.

links (200:1–32:1, ≤ 0.75 BPP), BigLILNet is the best method at every rate, surpassing even the road-specific SegRoadv2 and TopoRF-Net at a fraction of their parameter count. The margin widens as compression increases, reaching $+0.084$ IoU over the strongest baseline at 200:1. F1 score tracks closely with IoU across all operating points, so the method’s ranking and the crossover are robust to the choice of segmentation metric. This pattern is consistent with our hypothesis that the synthesis decoder, not latent quantisation, is the dominant source of degradation under aggressive compression: as the reconstruction becomes increasingly artefact-laden, the latent representation the baselines never see stays comparatively discriminative. Fig. 3 shows the effect directly, the latent models preserving connected road topology where the baselines fragment.

The advantage is not universal, and where it ends is itself informative. The crossover near 24:1–32:1 (Fig. 2) marks the point at which the full-resolution decoded image regains enough fine detail to overtake the $16\times$ -downsampled latent. Rather than a limitation, this is an actionable deployment

rule: it tells an onboard system which path to run for a given downlink budget, latent-domain below roughly 1 BPP, pixel-domain above, and pinpoints the bitrate at which the two pipelines are interchangeable.

Two further findings shape how such models should be built. First, BigLILNet improves on LILNet at every rate, so latent-domain accuracy is not yet bound by the compressed representation and continues to scale with capacity, which suggests headroom for larger latent-domain models to push the crossover toward lower compression. Second, both models interpolate smoothly to the unseen 0.25 and 0.75 BPP operating points, so a single network covers the codec’s rate range without per-rate retraining, a practical requirement for deployment against an evolving standard such as JPEG AI.

These gains come at a fraction of the cost: 4.3 M and 14.4 M parameters against 26–40 M for the baselines, with the synthesis decoder removed from the inference path entirely (the latencies in Table I exclude it and so understate the saving). For power and bandwidth-limited Earth observation platforms,

latent-domain inference is therefore most compelling precisely where conventional decode-then-infer pipelines struggle most.

V. CONCLUSION

In closing, we have shown that road segmentation can be performed directly on JPEG AI latents, and that doing so is most advantageous precisely where conventional decode-then-infer pipelines are weakest, under aggressive compression, while remaining far lighter than pixel-domain baselines. This extends compressed-domain inference from previously studied image-level recognition tasks to pixel-level prediction, showing that JPEG AI latents retain sufficient spatial structure for fine-grained analysis. In practice, this supports a bandwidth-adaptive pipeline that selects between latent- or pixel-domain inference based on available bandwidth or target accuracy.

Several directions open up from this work, road extraction is a deliberately demanding test of fine spatial structure; tasks that depend more on semantic content than on sharp boundaries, or that tolerate the latent's reduced spatial resolution, such as land-cover and cloud-cover mapping, change detection, or scene-level classification, may benefit even more from compressed-domain inference, and extending the evaluation to them would show how broadly the approach generalises. The consistent gain of BigLILNet over LILNet indicates that latent-domain accuracy is not yet capacity-bound: larger backbones, longer training schedules, and wider variance in training bitrates and codec operating points are a clear route to pushing the crossover toward lower compression and narrowing the remaining gap at high bitrates. Together these position latent-domain inference as a general foundation for efficient onboard analysis in bandwidth-constrained Earth observation, rather than a road-specific result.

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